

Delegation in Strategic Environments and Equilibrium Uniqueness

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- growing **antitrust concerns**
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 - ▶ EC Horizontal Guidelines (2023): algorithms as “a device to facilitate collusion (collusion by code)”
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Our paper: strategic environments where the concerns are moot, regardless of the intermediary details

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- two extremes of intermediary power:
 - ▶ **weak**: cannot punish deviators $\Rightarrow$ **coarse correlated equilibrium** (CCE)
 - ▶ **strong**: can coordinate adversarial punishments $\Rightarrow$ **individually rational correlated profiles** (IRCP)

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Main contribution: characterizations of games with a unique CCE or a unique IRCP

- uniqueness reduces to a **negative-sum externality** condition
- less demanding than expected, tractable, and suitable for collusion-proof design
- $\Rightarrow$ new sufficient conditions for uniqueness of NE and CE

Related literature

- **Correlated solution concepts:** CE (Aumann, 1974, 1987); CCE (Hannan, 1957; Moulin and Vial, 1978); IRCP (Myerson, 1991, Chapter 6)
- **Foundations:** communication $\rightarrow$ CE (Bárány, 1992; Ben-Porath, 1998; Gerardi, 2004; Lehrer and Sorin, 1997); regret learning $\rightarrow$ CCE, CE (Hannan, 1957; Foster and Vohra, 1997; Fudenberg and Levine, 1999; Hart and Mas-Colell, 2000); folk theorem logic $\rightarrow$ IRCP (Benoit, Krishna, et al., 1984; Fudenberg and Maskin, 1986; Aumann and Shapley, 1976);
- **Unique CE:** convex potential (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025); zero-sum (Forges, 1990; Viossat, 2006; Kattwinkel, Niemeyer, Preusser, and Winter, 2022; Kartik, Squintani, and Tinn, 2024); auctions (Feldman, Lucier, and Nisan, 2016)
- **Unique CCE:** strict social concavity (Even-Dar, Mansour, and Nadav, 2009; Hart and Mas-Colell, 2015); non-atomic congestion (Koessler et al., 2025); other examples (Nadav and Piliouras, 2010; Baudin, Scarsini, and Venel, 2023; Ahunbay and Bichler, 2025)

Roadmap

- ① Model & solution concepts: NE, CE, CCE, IRCP
- ② Unique IRCP
- ③ Unique pure CCE
- ④ Application: contests and potential games
- ⑤ Unique mixed CCE
- ⑥ Extensions and open problems

Model & solution concepts

Delegation as a correlated device

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Two extremes

weak intermediary: cannot punish vs **strong intermediary:** can coordinate adversarial punishments

CCE : weak intermediary

Coarse Correlated Equilibrium (CCE) (Hannan, 1957; Moulin and Vial, 1978)

a distribution $\mu \in \Delta(A)$ is a CCE if for every player i and fixed deviation $a'_i \in A_i$

$$\underbrace{\mathbb{E}_{\mu}[u_i(a)]}_{\text{utility of delegation}} \geq \underbrace{\mathbb{E}_{\mu}[u_i(a'_i, a_{-i})]}_{\text{utility of a fixed action } a'_i}$$

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contrast with Correlated Equilibrium (CE) ([Aumann, 1987](#)):

- CE captures recommendations rather than delegation
- a player can reconsider a recommendation a_i **after** seeing it
- μ is a CE iff for all i and maps $\sigma_i: A_i \rightarrow A_i$

$$\mathbb{E}_{\mu}[u_i(a)] \geq \mathbb{E}_{\mu}[u_i(\sigma_i(a_i), a_{-i})]$$

IRCP: strong intermediary

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- chain of inclusions:

$$\text{NE} \subseteq \text{CE} \subseteq \text{CCE} \subseteq \text{IRCP}$$

Unique IRCP

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Conclusion: in our search for games with a unique IRCP, we can focus on those with pure strict NE

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In fact, dominance is orthogonal to IRCP uniqueness

Example: parking enforcement

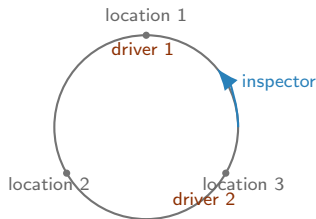
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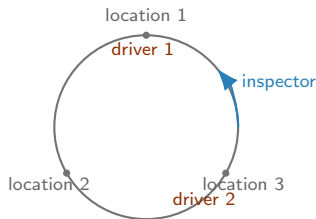
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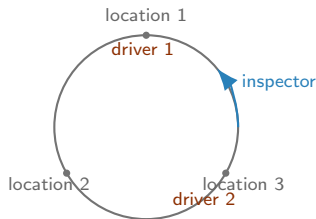


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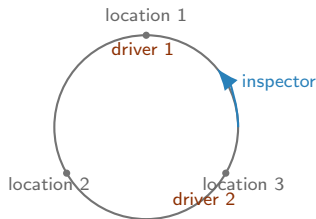


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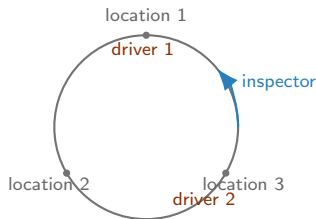


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- However, **parking legally** is a unique IRCP for $\frac{t}{2} > c$
- **Intuition:** at least one violator will be caught with probability $\geq \frac{1}{2} \Rightarrow$ prefers legal parking

Enforcement games

Enforcement game

A game $H = (A, v)$ is an **enforcement game** if there exists $a^* \in A$ such that

$$v_i(a^*) = 0, \quad v_i(a_i^*, a_{-i}) \geq 0 \quad \forall i, a_{-i}, \quad \sum_i v_i(a) < 0 \quad \forall a \neq a^*.$$

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Proof:

- if $\mu(A \setminus \{a^*\}) > 0$, then expected welfare < 0
- $\Rightarrow$ some player violates IR constraint



Characterization of games with unique IRCP

Theorem 1

A game G has a unique IRCP $\iff G$ is **affinely equivalent** to an enforcement game

- $G = (A, u)$ and $H = (A, v)$ are affine equivalent if $v_i = \gamma_i(u_i + \beta_i)$ with $\gamma_i > 0$ and $\beta_i \in \mathbb{R}$

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the dual to this LP gives γ_i and β_i

□

Unique CCE

Weak delegation: invariances matter

- CCE is invariant to payoff shifts that depend only on **others'** actions
 - ▶ $\Leftarrow$ CCE compares utility differences across own actions

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We first focus on pure CCE and discuss mixed CCE later

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- a connection between games with unique CCE and unique IRCP
- furthermore, a^* is the unique CCE of G iff a^* is the unique IRCP of the transformed game

$$v_i(a) = u_i(a) - u_i(a_i^*, a_{-i}),$$

that is, one can always take $\beta_i(a_{-i}) = -u_i(a_i^*, a_{-i})$ in strategic equivalence

Characterization of unique pure CCE

Theorem 2

A game G has a **unique and pure** CCE $\iff G$ is strategically equivalent to an enforcement game

- a connection between games with unique CCE and unique IRCP
- furthermore, a^* is the unique CCE of G iff a^* is the unique IRCP of the transformed game

$$v_i(a) = u_i(a) - u_i(a_i^*, a_{-i}),$$

that is, one can always take $\beta_i(a_{-i}) = -u_i(a_i^*, a_{-i})$ in strategic equivalence

- $\Rightarrow$ an easy-to check criterion for unique CCE

Testing for unique pure CCE

Criterion for uniqueness

a^* is a unique CCE iff there exist weights $\gamma_i > 0$ such that for all $a \neq a^*$,

$$\Phi_{a^*}(a) = \sum_i \gamma_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right) < 0$$

For symmetric games, can take $\gamma_i \equiv 1$

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- thus $\Phi_{a^*}(a_i, a_{-i}^*) < 0 \Leftrightarrow a^*$ is a strict Nash equilibrium
- uniqueness of CCE additionally requires **multilateral deviations** to be negative-sum in Φ

A simple sufficient condition for unique CCE

$$\Phi_{a^*}(a) = \sum_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right)$$

Idea: for strictly concave functions, coordinate-wise maximum $\Rightarrow$ global maximum

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- A_i is compact and convex
- each $u_i(a_i, a_{-i})$ is concave in a_i and convex in a_{-i}
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- this class includes linear Cournot and Tullock contests with strictly convex costs
- but there are economically relevant examples beyond

Application: contests & potential games

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- uniqueness of CCE $\Rightarrow$ no scope for collusion

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- choose weights $\gamma_i = 1/v_i$ in the local potential and use $p_1 + p_2 \equiv 1$ and get

$$\Phi_{a^*}(a) = \frac{1}{v_1} \left(u_1(a_1, a_2^*) - u_1(a^*) \right) + \frac{1}{v_2} \left(u_2(a_1^*, a_2) - u_2(a^*) \right)$$

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Remark: separable Φ is another class where coordinate-wise maximum $\Rightarrow$ global maximum

Ratio-based contests

- symmetric ratio-based contests: $p_1(a) = f\left(\frac{a_1}{a_2}\right)$ and $f(t) + f(1/t) = 1$
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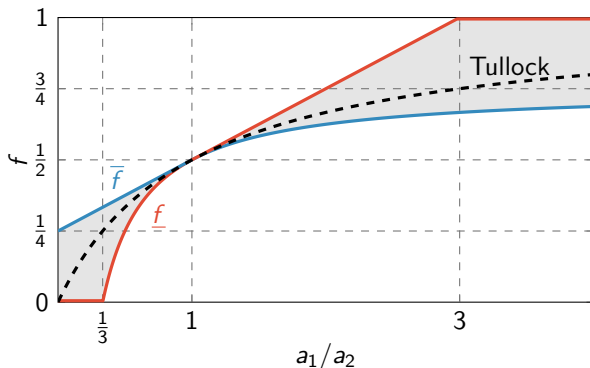
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- a contest is in this class $\Leftrightarrow f$ lies between $\bar{f}$ and $\underline{f}$



Potential games

- a **potential game** is one strategically equivalent to $u_i \equiv P$ (Monderer and Shapley, 1996)
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- separates environments with unique CE from those with unique CCE

Unique mixed CCE

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matching pennies		
	H	T
H	$(1, -1)$	$(-1, 1)$
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Conclusion: in contrast to IRCP, unique CCE can be mixed

Unique CCE is “almost pure”

Proposition

If $\text{CCE}(G)$ is unique, then either

- it is a **pure** NE, or
- exactly two players mix over two actions each, and the induced 2×2 game is of **matching-pennies type**.

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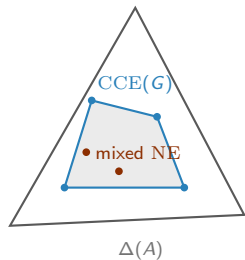
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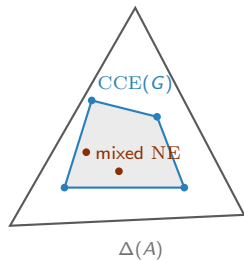
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 - mixed uniqueness is essentially a two-player artifact

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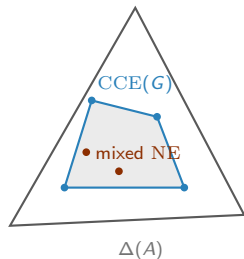
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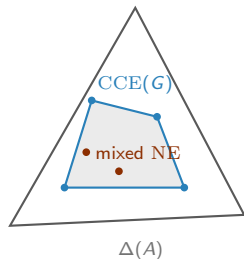
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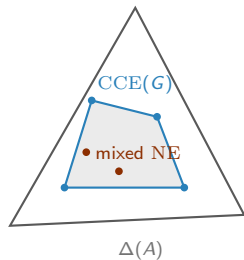
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Proposition

A quasi-strict NE is extreme in CCE $\iff$ it is pure or 2×2 mixing

Extensions

When does delegation just select among Nash?

- a more permissive benchmark for when delegation doesn't expand outcomes:

$$\text{delegation outcomes} = \text{conv}(\text{NE}(G))$$

- the intermediary only selects which NE to play

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- A mixed NE requires *indifferences* on its support, which cannot be generated as a convex combination of strict pure NE

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Open question

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- potential games: $\text{conv}(\text{NE}) = \text{CE}$ ([Neyman, 1997](#); [Ui, 2008](#); [Cao, Tan, and Zhou, 2025](#))
- zero-sum games: $\text{conv}(\text{NE}) = \text{CE}$ ([Forges, 1990](#); [Viossat, 2006](#))
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“No” in 2×2 games (Calvó-Armengol, 2006) and, more generally, two-person games (Cripps, 1995)

Take-away

- unique delegation outcomes in broad classes of games
- **strong intermediary (IRCP):** uniqueness $\Leftrightarrow$ enforcement up to affine equivalence
- **weak intermediary (CCE):** uniqueness $\Leftrightarrow$ enforcement up to strategic equivalence
- **design takeaway:** collusion-proofness is a tractable and permissive constraint
it is enough that multilateral deviations create **negative-sum externalities**

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Thank you!

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